

Nguyen Nhat Phat

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SUMMARY

Final-year Information Technology student (AI / Computer Vision concentration) with hands-on experience shipping edge-deployed computer vision systems and full-stack AI applications. Strong background in PyTorch, YOLO-based detection, and MLOps tooling (Docker, CUDA, Git).

EDUCATION

HCMC University of Technology and Engineering (HCM-UTE) Ho Chi Minh City, Vietnam
B.Eng. in Information Technology – AI & Computer Vision Concentration 2023 – Present

- GPA: 3.65 / 4.0
- HCM-UTE Talented Student Scholarship (2024); Academic Excellence Scholarship – Good Standing (2024)

EXPERIENCES

International Academic Exchange – BMW Program January 2026
University of Ulsan, South Korea

- Participated in an inbound academic exchange on Digital Twin technology and AI-based industrial simulation
- Collaborated with international students from South Korea and other countries on hands-on simulation sessions

PROJECTS

Adaptive Density-Based Geometry Compression for 3D Gaussian Splatting | *3D Reconstruction* 2026

- Developed a geometry-aware compression method reducing 3D Gaussian Splatting model size by **30%** with no quality loss, validated on 11 scenes across two benchmarks
- Accepted for publication at GTSD 2026 (Springer Track); received 1st Prize at Project Showcase 2026

Advanced Retinal Vessel Analysis Using Deep Learning | *PyTorch, CNN, Medical Image Analysis* 2025

- Led a deep-learning research project segmenting retinal blood vessels from high-resolution medical images
- Applied CNN architectures and image processing techniques for pixel-level segmentation accuracy
- Defended with a “Good” rating in the HCM-UTE university research program; presented at Euréka XXVII 2025 (VNU-HCM)

Real-Time Construction Site Safety Monitoring System | *YOLO, Edge AI, ONNX* 2026

- Led AI/ML development of a zone-aware PPE compliance detection system deployed on Jetson Xavier NX edge hardware
- Trained a YOLO26n-based model achieving **0.677 mAP@0.5 at 131 FPS** on embedded hardware
- Designed an edge-inference pipeline cutting network bandwidth usage by **99.93%** versus cloud-based streaming
- Awarded **3rd Prize** at the Edge Computing Innovation competition (HCM-UTE Faculty of Advanced Education)

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL
ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV
Tools & MLOps: Git, GitHub, Docker, CUDA, Kaggle, Google Colab
Practices: Agile / Scrum (certified), Product Ownership

AWARDS & CERTIFICATIONS

1st Prize – Project Showcase 2026, Academic Pathways Track (HCM-UTE)
3rd Prize – Edge Computing Innovation 2026 (Faculty of Advanced Education, HCM-UTE)
Outstanding Student in Scientific Research – FAEPRIME, HCM-UTE (2026)
Student of Five Merits – Ho Chi Minh City Level (2026) & University Level (2026)
Software Development with Scrum – Axon Active (2026)
IELTS 6.0